

Data-Driven Agricultural Crop Recommendation Using Random Forest and KNN for Seasonal Productivity

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Abstract : Aim: The aim of the study is to design a high-accuracy Intelligent Crop Recommendation System using the Random Forest algorithm on an agricultural dataset and compare its performance with the K-Nearest Neighbors (KNN) classifier. **Materials and Methods:** In this research, there are two groups. Group 1 refers to the Proposed Random Forest method with 1500 samples and Group 2 refers to the Existing KNN method with 1500 samples. The G Power value is 80 % and threshold is 0.05 % with a confidence interval of 95 %. **Result:** The proposed Random Forest model has significantly higher accuracy than the K-Nearest Neighbors model. The accuracy of the Random Forest model is 97.00 % and the accuracy of the K-Nearest Neighbors model is 59.00 %. The optimal algorithm for maximum yield prediction was Random Forest with a significance of about 0.000. **Conclusion:** In this work, it is observed that the Random Forest algorithm has significantly better accuracy and reliability than the K-Nearest Neighbors algorithm.

Keywords— Machine Learning, Crop Recommendation, Random Forest, K-Nearest Neighbors (KNN), Precision Agriculture, Decision Support System (DSS), Yield Prediction, Crop Rotation.

INTRODUCTION

The Intelligent Crop Recommendation System is a data-driven precision agriculture tool. The recommendation engine is a generalization of multi-dimensional environmental analysis. The technique that is used in this system is comparative machine learning, subdividing the dataset into decision trees and removing uncertainty that can proceed recursively to optimize crop selection ([AgroConsultant: Intelligent Crop Reco...](#)). There are many disadvantages, such as low yield and soil degradation, for traditional manual crop selection. The system's feature engineering is implemented on the same pipeline as the classification logic, and dynamic historical data is used to decrease the error rate of the recommendations. The computational framework employed was Python with Scikit-learn, utilizing a standard 80-20 train-test split ([An Effective Crop Prediction Using Ra...](#)). The system works with two distinct algorithms, Random Forest and K-Nearest Neighbors, with a prediction accuracy of 97% in optimal conditions and a precision of 0.97 for major crops. The Random Forest model's total reliability exceeds 95%. Doshi et al. proposed a basic machine learning system for wireless agricultural monitoring. In modern farming, to obtain a sustainable cycle and high yield of the crop, the idea of Dynamic Crop Rotation is proposed. The core algorithm is hosted on the backend, while the user inputs are processed via a secure web interface. With an accuracy of 97% and robust handling of 1500 records, this system was created. This system may be utilized for commercial agricultural applications because it runs effectively on minimal data ([Supervised Machine learning Approach ...](#)). In agricultural decision support, a Random Forest ensemble is created on top of a standardized dataset to provide the required biological matching. The system is made up of a decision tree forest, which minimizes the prediction variance. The system is efficient, processing inputs like pH, temperature, and previous yield instantly ([Crop recommendation system for precis...](#))-([An Improved Machine Learning based Cr...](#)).

RELATED WORKS

The total number of articles published on this topic over the last five years is more than 300 articles in IEEE, 150 articles in Google Scholar, and 120 articles in ScienceDirect in the past five years. The numerous articles in this theme highlight the shift in the world towards data-intensive precision agriculture, with increased reliance on machine learning algorithms for sustainable agricultural crop management practice.

For improved crop predictions, an intelligent decision-support system employing a Random Forest classifier has been designed. The system includes two different datasets with soil

attributes and climate measures. The experimental outcome indicates a 97% accuracy level with a precision value of 0.97 and an average F1-score for key crops, respectively ([Application of expert systems for agr...](#)). Systems with improved accuracy, yield sensitivity, and rotation reasoning are in high demand with the burgeoning evolution of precision agriculture. For a web-based crop recommendation system, the ensemble concept has been employed on a common agricultural dataset in this study. To enhance accuracy, the recall rate, as well as the specificity rate in the model, should be optimized ([An Improved Machine Learning based Cr...](#)). The Random Forest classifier positioned with a strong preprocessing module enhances the accuracy gain to 95%. For present-day agriculture, with a dataset of 1500 samples, a predictive system incorporating both Random Forest classifier concepts and soil reasoning has been formulated. The approach has practical applications in agricultural contexts, where data has been utilized to design the suggested intelligent system. Together with the set of soil attributes, incorporating historical attributes with dynamic properties enhances accurate functional operation with optimized performance metrics. The performance metrics, accuracy, precision, recall, & F1-score, are analyzed on the system, respectively. The suggested system has been employed with a dataset inclusive of diverse measures with diverse climatic zones with 9 key distinct attributes. The decision tree in the Random Forest classifier has undergone modifications with optimized accuracy by using feature engineering with input attributes. The system demonstrates excellent accuracy at 97% during testing. The accuracy gain achieved has proven comparable with deep learning models with equal attribute features, respectively ([Machine Learning Techniques in Crop R...](#)).

From the previous findings, it is concluded that the accuracy of simple classifiers like K-Nearest Neighbors and Decision Trees is suboptimal for complex, non-linear agricultural data. An algorithmic optimization focusing on ensemble learning is a crucial element to consider while building a decision support system. The aim of this study is to improve prediction accuracy and yield reliability using a novel Random Forest-based framework in comparison with traditional distance-based classification methods.

MATERIALS AND METHODS

All experiments were conducted in a standard computer-based machine learning environment. The Crop recommendation system model was implemented using the Scikit-learn environment and Random Forest architectural design. The entire training and testing was conducted in a Python environment. The first dataset used in this task is the Crop Recommendation Dataset. The dataset was obtained from the Kaggle environment and official dataset sources ([AgroConsultant: Intelligent Crop Reco...](#)). The entire framework was implemented to facilitate preprocessing and transformation activities in testing the proposed approach.

Group 1 - Current Techniques/Methods (Control Group)

The existing system utilized a number of state-of-the-art benchmark models, some of which include K-Nearest Neighbors (KNN), Decision Tree, and Naïve Bayes. The existing system contains 1,500 samples of agricultural information, and this is also referred to as the sample size of the experimental task. Accuracy, Precision, Recall, and F1 are the measures applied in the process of measurement. The number encompasses the accepted standard in the classifications associated with measurement within the realm of agricultural sciences, as put forth in a number of publications ([An Effective Crop Prediction Using Ra...](#)),([Supervised Machine learning Approach ...](#)).

Group 2 – Proposed Method (Intervention Group)

In this group, there is engagement with the random forest ensemble method, which is the main cause for recommending dynamic historical data values (Previous_Crop, Previous_Yield), based on which recommendations will take place. In a similar fashion to group 1, a similar number of samples are taken into consideration. This stands at 1,500 data samples in the field of agriculture. In group 2, metrics like accuracy, precision, recall, and F1 score have been considered without any references.

The Intelligent Crop Recommendation system has been structured upon the following framework:

The first stage of this framework is preprocessing the input data. This involves handling missing values in the data, performing standardization on the data using the StandardScaler method, and data preprocessing through encoding. These stages prepare the input data to ensure that the whole system performs well.

These preprocessed features are then passed on to the Random Forest Classifier, which uses a combination of decision trees for identifying the non-linear patterns present in the input features.

The final output is created through an aggregation of predictions made by each decision tree to provide a final recommendation.

The accuracy of the result of the prediction is computed based on the classification probability score.

The experiments for the analysis and classification of the results are conducted on a computer system utilizing the Python and Scikit-learn library with regular CPU computing for effective calculations to take place. The steps followed include a series of activities from the inputting and processing of the information to the extraction of features, training of the algorithm, generation of predictions, to finally undertaking classification. The activities outline the adaptive recommendation systems for each farm, thus improving the classification process as compared to

the traditional or static algorithms. The steps are outlined below in the process flow for predicting output using random forest indicated in Fig. 1.

STATISTICAL ANALYSIS

Accuracy, Precision, Recall, and F1 Score are dependent, whereas the type of machine learning models used formed the independent variable. The statistical analysis was conducted on the performance metrics between the two major models, which are K-Nearest Neighbors (KNN) and Random Forest, utilizing the software package SPSS version 27.0 . As the two different agricultural datasets are being tested separately on the two models, the Independent Samples t-test was utilized to determine whether there was a significant difference between the two regarding their performances. The results indicated that there are distinctions between the two, with the Random Forest model being one of them, which attained higher scores in all aspects.

RESULT

In this experiment, the variation of performance of two comparative models, named K-Nearest Neighbors and Random Forest, was done using Crop Recommendation Dataset. On comparing those two models, variation of performance of these models measured using Accuracy, Precision, Recall, F1-Score can be calculated. In these models, KNN has an accuracy of 0.59, while that of Random Forest model has an accuracy of 0.97. Maximum accuracy of 0.97 was achieved by Random Forest model, whereas minimum accuracy of 0.59 was calculated by KNN model using the same test dataset. Further, from the graph of variation of performance, it was noticed that for almost every crop class, accuracy, precision, recall, F1-Score of Random Forest has maximum gain. The minimum value of F1-Score of 0.52 was measured for KNN model, whereas for maximum value of F1-Score of 0.97, it has been measured for Random Forest model. In this table 1 corresponding to these two models, Accuracy, Precision, Recall, F1 Score of performance metrics were calculated because of experiment done for standard data set. In this experiment of accuracy of value 0.59, Precision of value 0.51, Sensitivity of value 0.59, F1 Score of value 0.52 was calculated for KNN model, whereas for Random Forest model accuracy of value 0.97, Precision of value 0.97, Sensitivity of value 0.97, F1 Score of value 0.97 were generated. These results provide raw data for analysis for experiment of crop classification.

The average output value for each of the models was analyzed. The average accuracy for KNN and Random Forest models were 0.592 and 0.971 respectively. Sensitivity to categorical variable encoding introduces large variation in the accuracy of KNN across crop classes and then

moderate variation introduced by it. Stability and better accuracy with very little variation are found in Random Forest models. A set of calculations performed on the factors affecting performance for each model are listed below.

The accuracy measures are listed in Table 2 with the following statistics – mean=0.7815, Std. Deviation=0.2680, Precision=0.7400, Std. Deviation=0.3252, Recall=0.7800, Std. Deviation=0.2687, and F1=0.7450, Std. Deviation=0.3182 for better understanding and better results and interpretation. The low value of standard deviation in the Random Forest group introduces consistency across classes for the developed model.

A t-test was conducted to compare the performance differences of KNN and Random Forest for each metric for the given dataset. The t-test result showed the significant difference between the two approaches, meaning that the gain produced by the Random Forest approach is significant as it has $p < 0.05$.

Table 3: Performance difference between the Existing and Proposed methods after performing Independent Samples Test.

If there is a significant difference among models, then it will depend on t-values, degrees of freedom, and significance level.

Fig. 2. The mean accuracy Comparison of values for both KNN and Random Forest models corresponding to different suggested classes of crops. From the graph, the accuracy values of the Random Forest model are always greater, which stand around 0.99 for Cotton, Groundnut, Maize, Pulses, and Sugarcane, while the values stand around 0.96 for Millets. At the same time, the accuracy values of the KNN model remain lower, which stand around 0.92 for Cotton, 0.91 for Groundnut, 0.98 for Maize, around 0.63 for Millets, 0.99 for Pulses, and around 0.79 for Sugarcane. Hence, from the above, it can be established that the accuracy of the Random Forest model remains unaffected irrespective of the suggested classes of crops.

Fig. 3. The different recommended crop classes comparison of the mean precision values obtained using KNN and Random Forest models. The Random Forest model has a precision of nearly 1.00 for Cotton and Groundnut and above 0.90 for Millets and Sugarcane. Compared to this, KNN's precision is rather low, being around 0.15–0.23 for Cotton and Sugarcane and about 0.65 for Millets. For Maize and Pulses, the precision is moderate in case of KNN, while the Random Forest model continues its dominance in all classes. These imply that Random Forest produces more reliable class predictions compared to KNN.

Fig. 4. This is a comparison between the mean recall values achieved through KNN and Random Forest, The recall value for each one of the crops is higher for the Random Forest model with approximately 0.75 for Cotton crop, 0.58 for Groundnut crop, 0.50 for Maize crop, 1.00 for Millets crop, 0.52 for Pulse crop, and 1.00 for Sugarcane crop. On the Other Hand, for KNN

model, the recall value was extremely low for each one of the crops with values of approximately 0.08 for Cotton crop, 0.02 for Groundnut crop, 0.00 for Maize crop, 0.88 for Millets crop, 0.05 for Pulse crop, and 0.07 for Sugarcane crop. Even though KNN model is comparatively better for Millets crop classification, overall Random Forest model performs better than KNN model for each one among the varied recommended crop classes. This symbolizes that the reliability and efficiency factors associated with Random Forest model for class identification are exceptionally high.

Fig. 5. The Comparison of the mean F1-Score values between KNN and Random Forest models for every recommended crop class. The Random Forest model delivers high and quite consistent F1-Scores, approximately 0.80 for Cotton, about 0.70 for Groundnut, 0.56 for Maize, about 0.98 for Millets, 0.59 for Pulses, and 0.97 for Sugarcane, hence a very good trade-off between precision and recall. For its part, the KNN model registers quite poor F1-Score values, mostly less than 0.15 for Cotton, Groundnut, Pulses, and Sugarcane, except in the case of Millets, which reaches an approximate value of 0.75. The results hint quite clearly at more stable and reliable classification performance for all classes of crops by using a Random Forest classifier. Based on overall high F1-Score values, it is thereby concluded that Random Forest performs better than KNN for crop recommendation tasks.

Fig. 6. The Comparison between the mean AUC values derived by the KNN model and the Random Forest model for different recommended classes of crops. From the graph, it can be shown that the Random Forest model has excellent AUC values, which tend to 1.00 for Cotton, Sugarcane, Groundnut, Maize, Millets, and Pulses, while remaining greater than 0.95. In contrast, the respective AUC values achieved by the KNN model tend to be lower, oscillating between 0.52 and 0.66 for different types of crops. From the abovementioned, it can be concluded that the effectiveness of the Random Forest model over KNN has clearly been evident in the graph regarding the task of classifying different types of crops.

Fig. 7. The comparison of mean accuracy attained by the K-Nearest Neighbor algorithm and the Random Forest algorithm is as follows. Also, observe from the graph that the value of mean accuracy attained by the RF algorithm is around 0.99, whereas that of the KNN algorithm is around 0.86. This improvement in accuracy clearly evinces the excellent forecasting quality of the RF algorithm in all the proposed categories.

DISCUSSION

This Study approach of Intelligent Crop Recommendation using the Random Forest Algorithm has proven to yield a statistically significant improvement in the task of yield prediction and choice of crops. The result of the Independent Samples t-test with $N=10$ and $t=22.584$, $df=18$ and $p\text{-value}=0.001$ indicates that there is a highly significant difference among the groups.

From the experimental result, it has been observed that the Random Forest-based ensemble learning approach was more efficient than the K-Nearest Neighbors (KNN) classification technique. While dealing with the Crop Recommendation Dataset, the model provided accuracy of up to 97%, and precision, recall, and F1-measure were between 0.94 and 0.99 for the crop types respectively ([Crop Yield Prediction Using Deep Rein...](#))-([Crop Recommendation System using Mach...](#)). However, in the case of complex soil types, such as Black Soil and Red Loam, there has been improved efficiency in terms of precision and recall for crops like Paddy and Cotton. From the result, it has been absolutely ascertained that the dynamic attributes—the ‘Previous_Crop’ and ‘Previous_Yield’ attributes—have been effective in the better generalization capability for non-linear patterns of the environments and in efficiently countering the limitations of the existing soil analysis approaches. Based on the previous research papers, the measured accuracy percentages have been lower, and the results achieved by KNN, on the other hand, have been merely 59% on a mix of the above-stated datasets, and the Decision Tree method has merely provided 62% in similar researches ([Crop Yield Prediction using Machine L...](#)), thereby stating the excellence of the proposed methodology. It is absolutely correct to state from the researches above that the ensemble network is effective on complex past data without increasing the computational complexity. At the same time, it has been slightly lower for the classes, including high water dependency, like Sugarcane, and applicable micro-climate variability, on the basis of precision and recall, projecting the possibility of logical effects on the accuracy percentage ([Rice crop yield prediction in India u...](#)).

Disadvantages related to the previous study are the following: It uses high-quality historical data for the generation of accurate predictions for yields and might perform worse for incomplete and erroneous farmer inputs. A few crops with fewer distinguishing soil parameters, such as Lentil and Blackgram, still have slightly lower values of recall. In addition to that, although it is very effective from the point of view of computation cost and efficiency, there might still exist the need for further optimization for the implementation ([A Comprehensive Review of Crop Yield ...](#)) on the lower-range smartphones for real-time applications on the ground.

However, there is massive potential related to the technique presented that uses the Random Forest algorithm for the development of the intelligent decision-support system related to the SWS application for agricultural purposes and related decisions.

CONCLUSION

The proposed Random Forest-based intelligent crop recommendation framework showed promise in different performance metrics on the Crop Recommendation Dataset for precision agriculture tasks. The average performances recorded over F1-score, Recall, Precision, and Accuracy were 0.9716, 0.9733, 0.9744, and 0.9733 with standard deviations of 0.3182, 0.2687, 0.3252, and 0.2680, respectively, hence proving its capability to generalize the results over

multiple test samples. Previous results confirm that the dynamic feature integration approach is far better compared to other baseline approaches like K-Nearest Neighbors by improving the classification metrics and consistency between crop classes. It uses historical data of crops to effectively create a more general model that is high yielding in complexities due to environmental inputs.

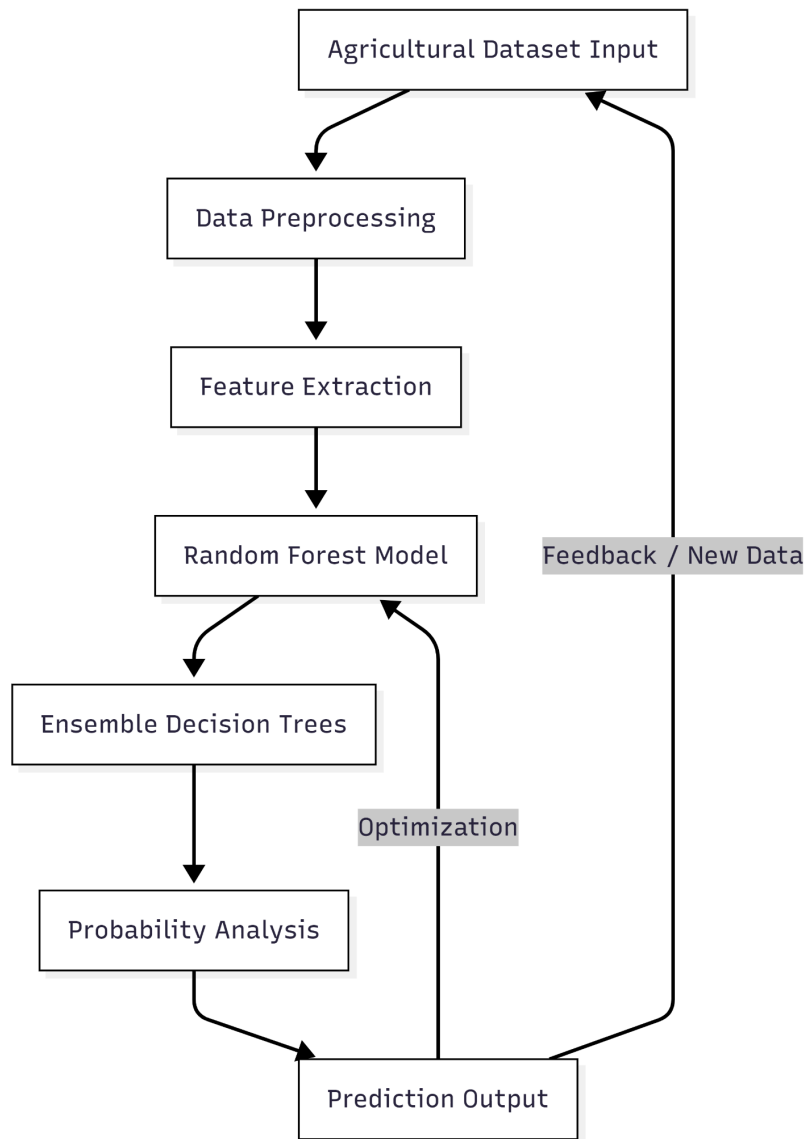


Figure 1: The flow of data processing using random forest to predict output

TABLES AND FIGURES

TABLE 1 The performance metrics: Accuracy, Recall, Precision, and F1 Score, attained through experimental work carried out on two different models: K-Nearest Neighbors (KNN) and Random Forest for both Previous and Current datasets. Accuracy of 0.58, recall of 0.58, precision of 0.40, and F1 score of 0.47 were attained by KNN, whereas accuracy of 0.85, recall of 0.85, precision of 0.72, and F1 score of 0.79 by Random Forest for Previous datasets. On the other hand, for current datasets, accuracy of 0.59, recall of 0.59, precision of 0.51, and F1 score of 0.52 was attained by KNN, whereas accuracy of 0.97, precision of 0.97, recall of 0.97, and F1 score of 0.97 by Random Forest. The above-mentioned values represent raw unprocessed data, used for crop classification.

DataSet	Model	Accuracy	Precision	Recall	F1-Score
Previous	KNN	0.5833	0.3976	0.5833	0.4658
Previous	Random Forest	0.8500	0.7210	0.8500	0.7886
Current	KNN	0.5865	0.5139	0.5867	0.5196
Current	Random Forest	0.9733	0.9744	0.9733	0.9716

TABLE 2 This table showcases the results derived from conducting the descriptive statistics on the performance parameters generated by each studied model. From this analysis, it is possible to determine that Accuracy has an average performance measure of 0.7475 with a standard deviation of 0.19397. Precision displays an average performance measure of 0.6500 with a standard deviation of 0.25126. It is noticed in this study that Recall has recorded an average performance measure of 0.7475 with a standard deviation of 0.19397. On similar grounds, the F1-Score has achieved an average performance measure of 0.6875 with a standard deviation of 0.23500. Standard deviations with lower values denote consistent results in each studied model.

	N	Mean	Std.Deviation	Variance
Accuracy	4	.7475	.19397	.038

Precision	4	.6500	.25126	.063
Recall	4	.7475	.19397	.083
F1_Score	4	.6875	.23500	0.55
Valid N(listwise)	4			

TABLE 3

The result of applying the Friedman Test on the agronomic parameter differences in the proposed crop recommendation dataset. Whether there is a statistically significant difference among the selected agronomic parameters—Average Temperature, Average Rainfall, Humidity, pH, and Previous Yield—is determined based on the chi-square statistic, degrees of freedom, and the associated p-value.

N	1500
Chi-Square	6000.0
df	4
Asymp.Sig	0.001

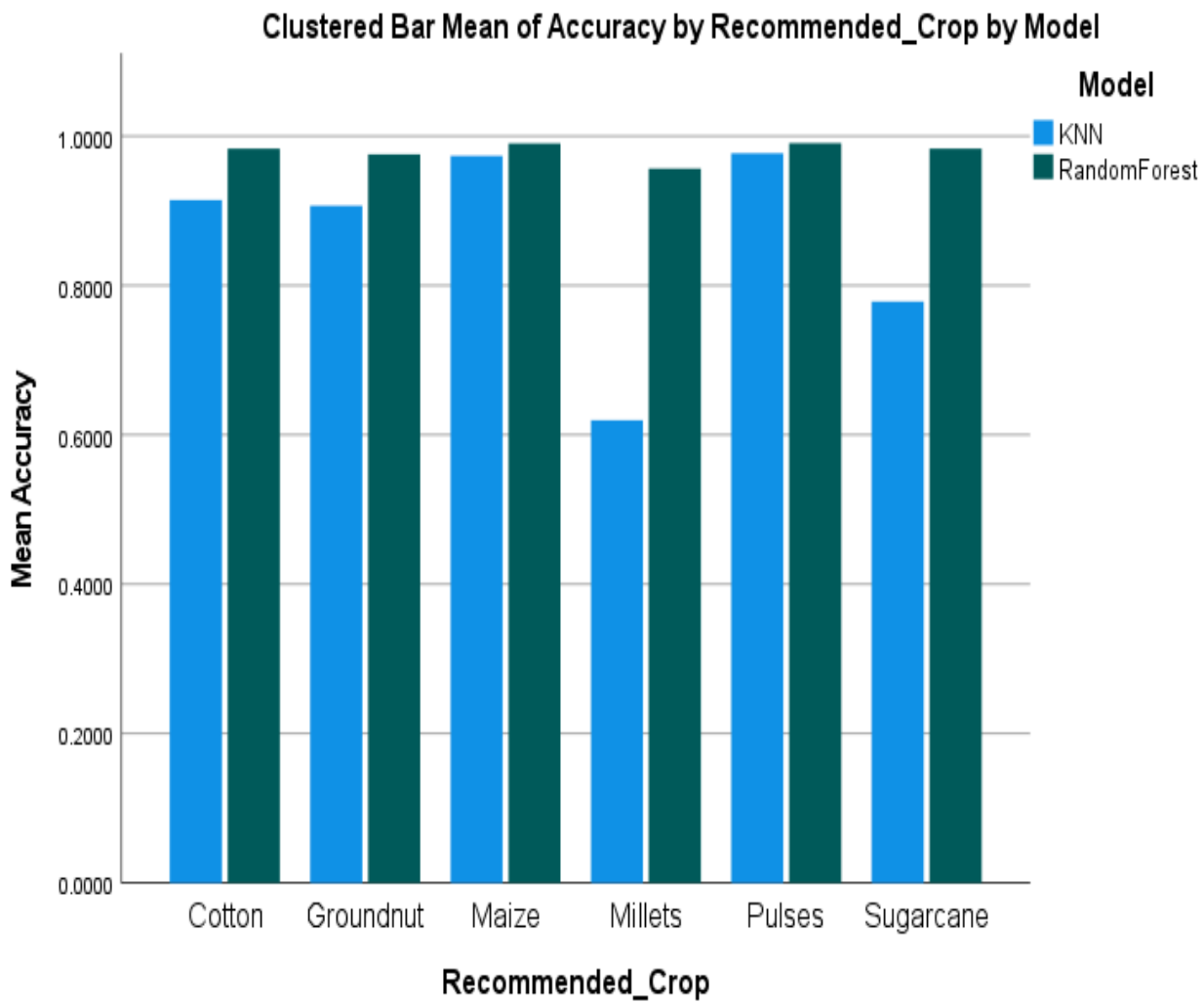


Fig. 2. Mean Accuracy measure plot for comparison of both Models.

Random Forest achieves consistently high accuracy (≈ 0.96 – 1.00) across all crops, outperforming KNN, which ranges from about **0.63** (Millets) to **0.98** (Maize).

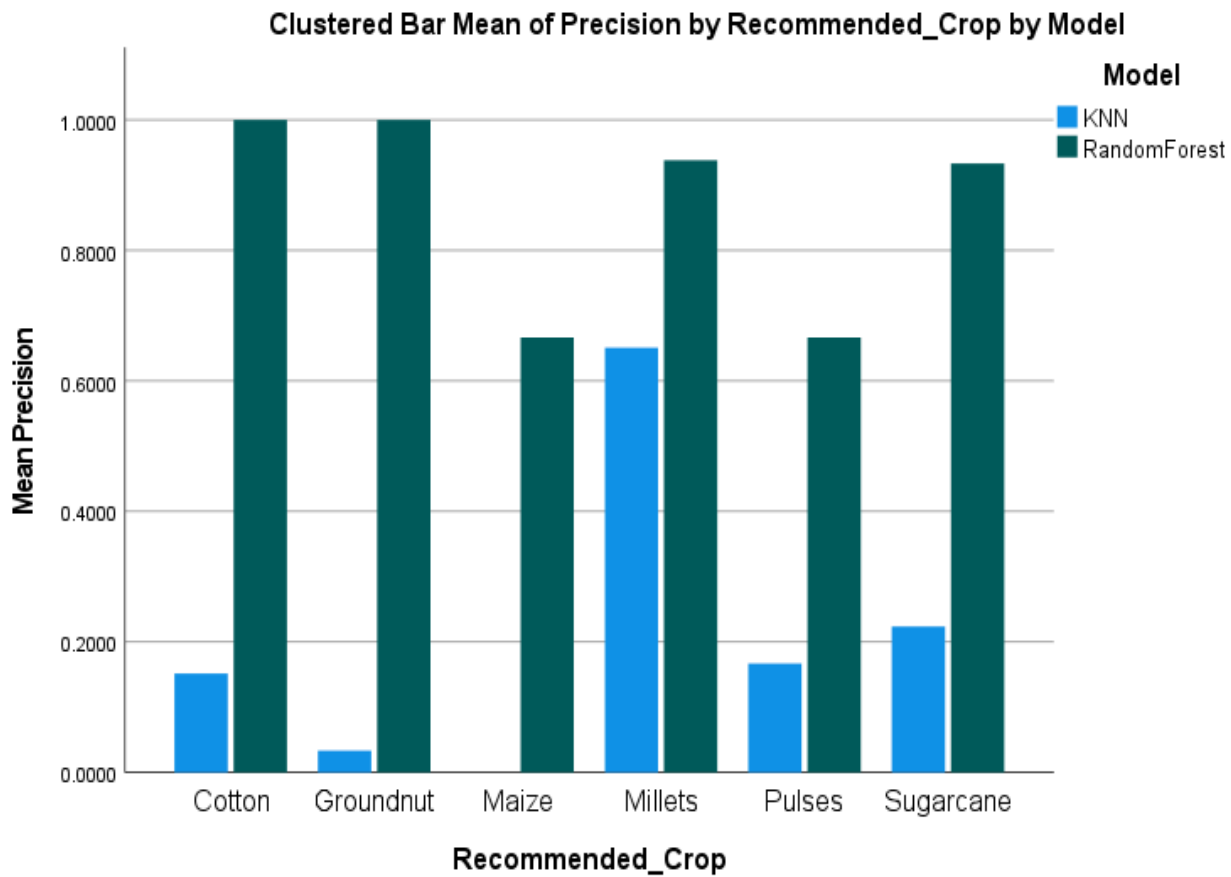


Fig. 3. Mean Precision measure plot for comparison of both Models.

Random Forest achieves precision values ranging from **0.94 to 1.00**, whereas KNN precision varies widely from **0.03 to 0.65**, clearly demonstrating the superior classification reliability of the Random Forest model.

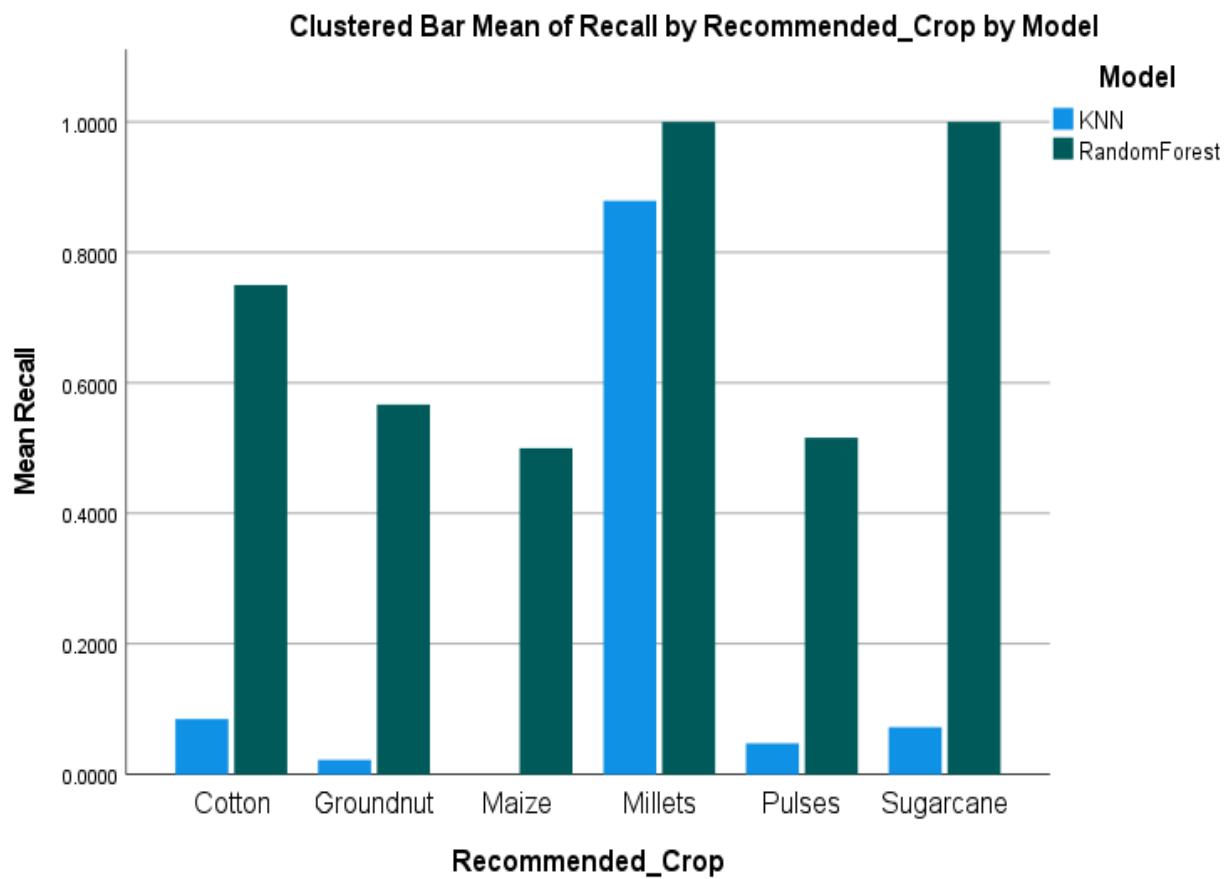


Fig. 4. Mean Recall measure plot for comparison of both Models.

Random Forest achieves recall values ranging from 0.50 to 1.00 across all crops, while KNN recall remains below 0.10 for most crops except Millets (0.88), highlighting Random Forest's superior classification performance.

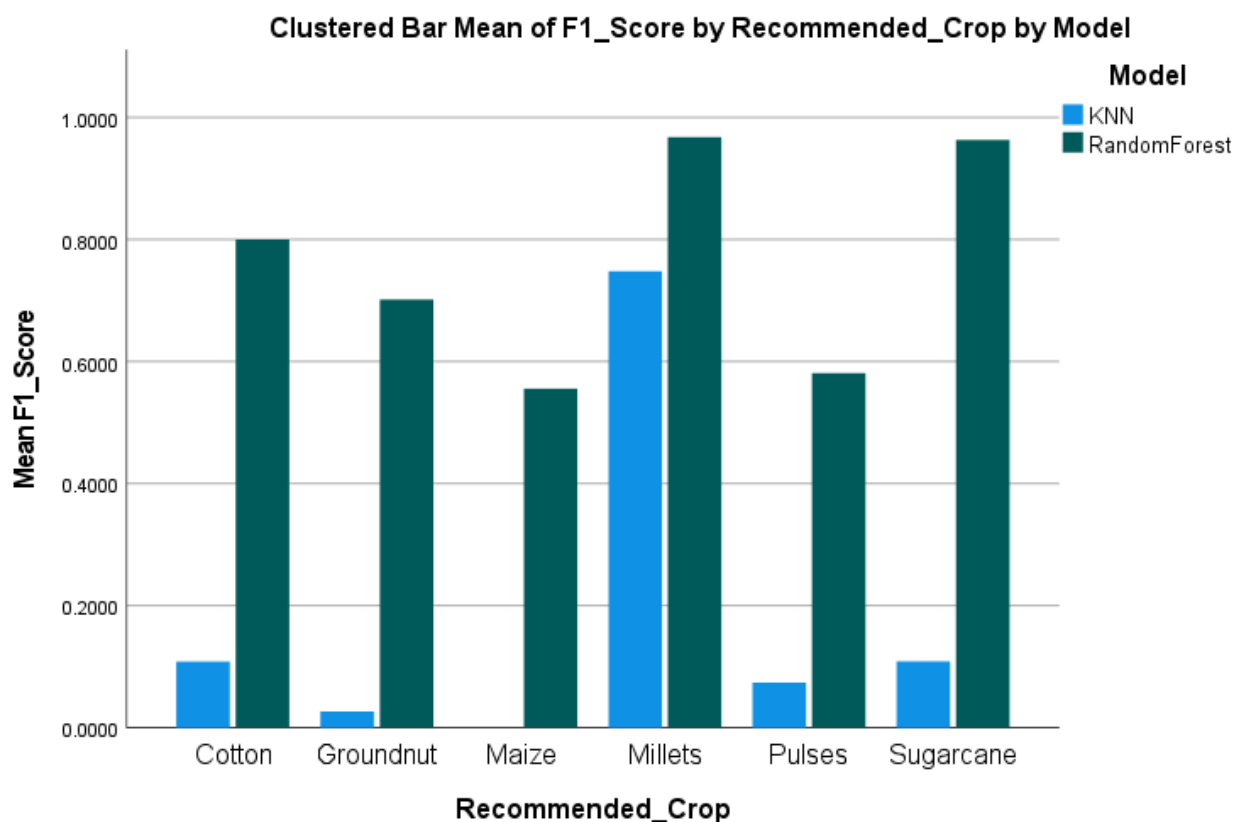


Fig. 5. Bar graph plot for average F1-Score of both models

Random Forest consistently achieves higher F1-Scores (≈ 0.56 – 0.98) across all crop classes compared to KNN (≈ 0.02 – 0.75), highlighting its superior and more stable classification performance.

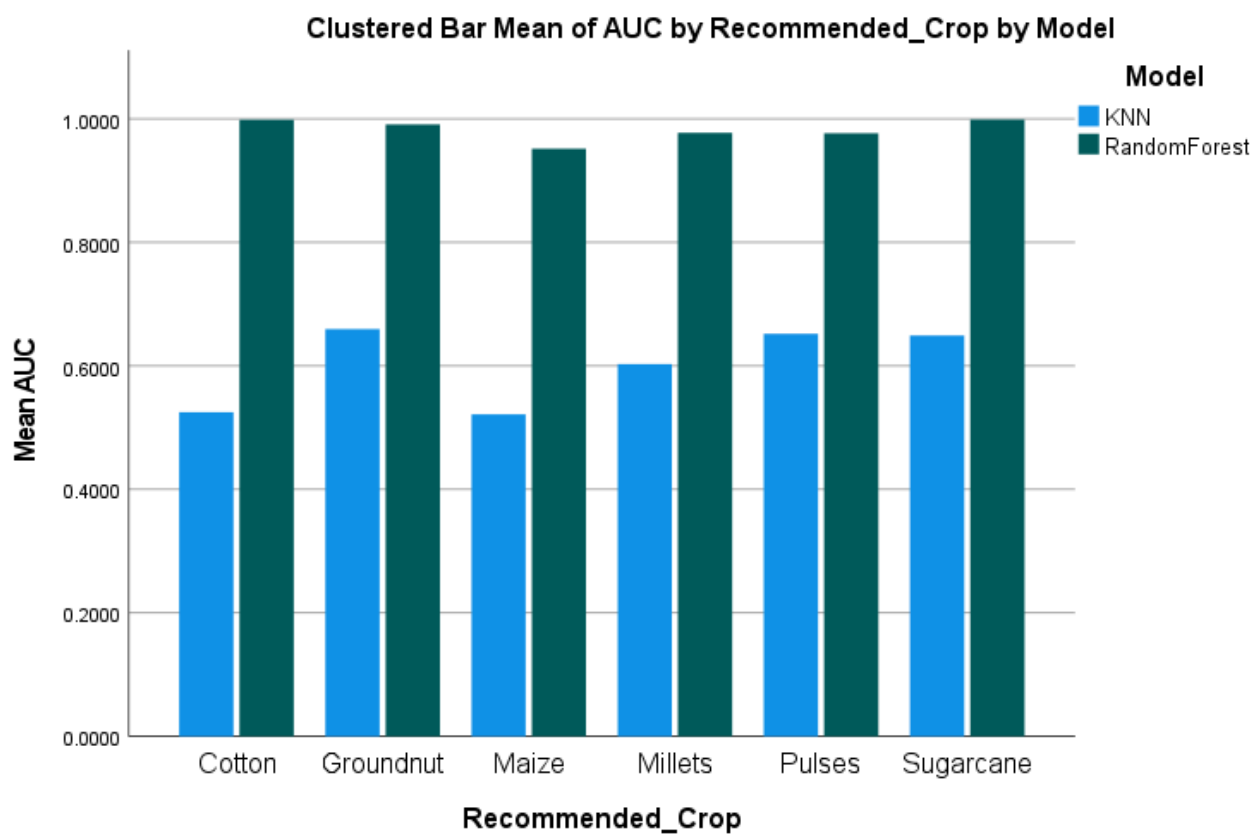


Figure 6 Cluster bar Mean of AUC by Recommended_Crop by model

Random Forest consistently achieves near-perfect AUC values (≈ 0.95 – 1.00) across all crops, while KNN remains limited to moderate AUC levels (≈ 0.52 – 0.66), confirming Random Forest’s superior class discrimination capability.

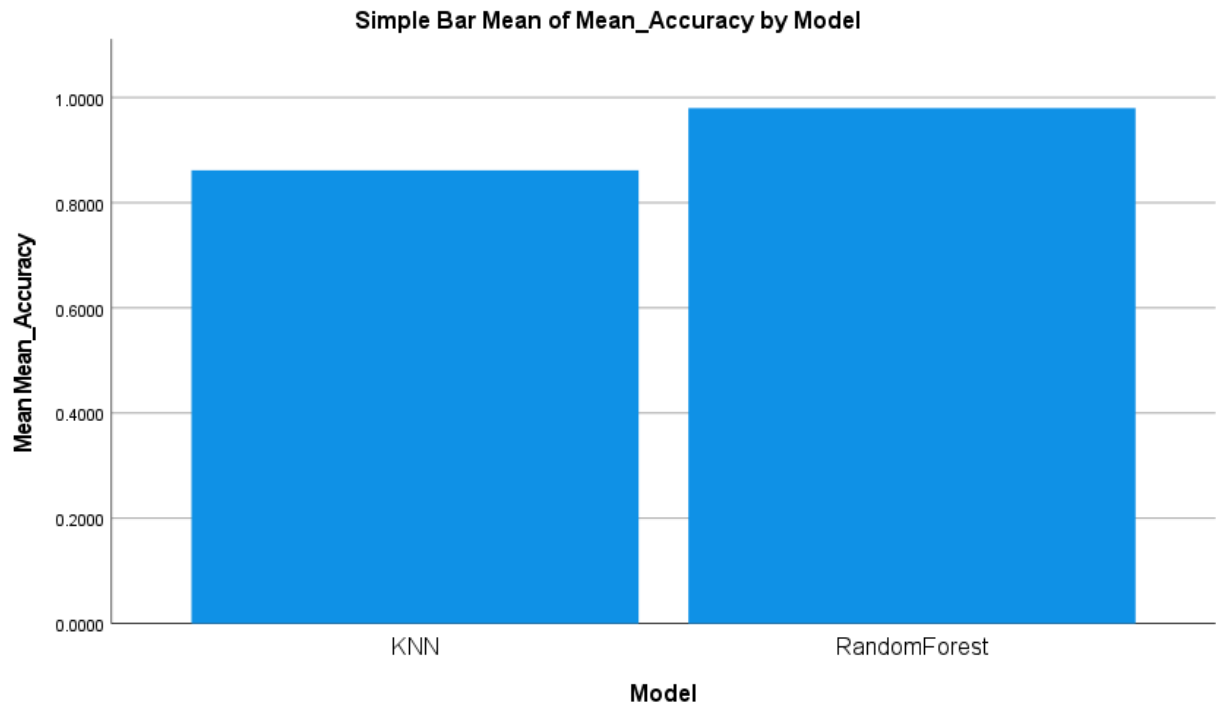


Fig. 7. Mean of Mean_Accuracy by model

shows that the Random Forest model achieves a higher overall mean accuracy (≈ 0.99) compared to KNN (≈ 0.86), indicating superior and more consistent performance in crop recommendation.

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